

Hossam Abdelhakim

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SUMMARY

Software Engineer with 3+ years of experience in **C++**, **Python**, and **JavaScript**, specializing in **backend development** and **system optimization**. Ranked in the **top 0.4% on LeetCode** with a **1000+ day streak**. Founded **MaTb3aa**, mentoring 3000+ students and coaching 7 university teams to top **ACM ICPC** rankings. Adept at building **scalable, high-performance systems** and delivering **AI-driven solutions** for complex technical challenges.

WORK EXPERIENCE

Tahaluf AI Emarat Technical Solutions

C++ Software Engineer / AI RD Team

Abu Dhabi, UAE

Apr 2023 – Dec 2024

- Developed **scalable C++ solutions** for intelligent traffic management systems, improving **real-time data handling by 23%**.
- Enhanced **system responsiveness by 40%** (from 500ms to 300ms) through caching and optimization, reducing latency for mission-critical operations.
- Designed and implemented a **distributed architecture** for TRAFFITIX-Managers, improving **system resilience by 20%** and ensuring fault tolerance.
- Led robustness testing across projects, enhancing stability and reducing failure points by 15%.
- Collaborated with cross-functional teams to integrate solutions and conducted code reviews, ensuring seamless project delivery.
- Technologies:** C++17/20, CMake, Multithreading, Caching Systems, Distributed Systems, Real-time Processing

Upwork

Software Engineer

Remote

Mar 2023 – Nov 2024

- Delivered high-stakes coding solutions across **446+ billable hours**, generating **\$16.3K+ in revenue** at \$35/hr with a **100% Job Success Score**.
- Architected an **AI-driven code generation system** using **Python** and **TensorFlow** for the *Talent Store: Coding Rapid* project, earning a perfect **5.0 client rating** and **\$3.3K+ in revenue**.
- Reduced client runtime complexity from **$O(n^2)$ to $O(n \log n)$** across 15+ critical-path algorithms, significantly improving application performance.
- Earned **Upwork Top Rated Plus Badge (2023)** and maintained a **5.0/5.0 Client Satisfaction Score**.
- Technologies:** C++, Python, Algorithms, Data Structures, AI/ML Integration, TensorFlow, Performance Optimization

Eatura

Backend Engineer

Cairo, Egypt

Sep 2021 – Jan 2023

- Developed **RESTful APIs** using **Node.js/Express.js** to connect **200+ suppliers** with distributors, handling authentication, order processing, and inventory management with **MongoDB**.
- Optimized **MongoDB query performance by 55%** through indexing and aggregation pipelines, enabling high-frequency inventory updates for large-scale operations.
- Collaborated with frontend and product teams to deliver seamless API integrations, enhancing supplier satisfaction and operational efficiency.
- **Technologies:** Node.js, Express.js, MongoDB, RESTful APIs, Microservices, JavaScript/TypeScript

EDUCATION

Bachelor of Computer Science
Aug 2017 – June 2021

Assiut, Egypt

- Awarded **Certificate of Excellence** for exceptional performance.

PROJECTS

Surgical Simulation Engine (C++)

- Enhanced surgical precision for bone transplants, impacting global healthcare outcomes.
- Developed a high-performance simulation engine using **C++**, focusing on real-time processing and accuracy.

Real-time Chat Application (Node.js, WebSocket)

- Built a scalable chat app handling **1K+ concurrent users** with **WebSocket** for real-time messaging.
- Implemented **load balancing** and **caching** to ensure low latency and high availability.

Arabic Language Compiler (C#, Parsing)

- Created a custom scanner/parser for Arabic linguistic structures, reducing processing time by **40%**.
- Leveraged **C#** for compiler development and advanced parsing techniques.

SKILLS

- **Programming Languages:** C++14/20, Python, JavaScript, Java, Go
- **Backend Technologies:** Node.js, Express.js, RESTful APIs, Microservices
- **DevOps & Cloud:** Docker, Kubernetes, Terraform, Jenkins, AWS, Azure, Grafana, CI/CD, GitHub/GitLab
- **Databases:** MongoDB, MySQL, SQLite, Redis, GraphQL, Query Optimization
- **Software Engineering:** OOP, SOLID Principles, Design Patterns, Architecture
- **Algorithms:** Dynamic Programming, Graph Theory, Trie Structures
- **Tools:** Git, CMake, Visual Studio, Eclipse, IntelliJ
- **Soft Skills:** Leadership, Mentorship, Communication, Problem-Solving

EXTRACURRICULAR ACTIVITIES

- **Coach & Founder, ICPC BUA & EELU Q Community:** Mentored **3000+** students, leading teams to top placements in **ECPC/ACPC**.
- **Technical Educator, YouTube (7.8K+ subs):** Created **400+** videos on C++, algorithms, and FAANG interview prep.
- **Coach & Judge, Egyptian Olympiad in Informatics:** Coached students to win **1 gold, 2 silver, and 5 bronze medals**.
- **Speaker, University Tech Events:** Delivered **25+** workshops on problem-solving and C++ at 7 institutions.

CERTIFICATIONS

- ACPC Finalist
- Finalist, 2nd place in Upper Egypt
- ACM IEEE Minia Finalist, 4th place
- 1100+ Daily Streak on LeetCode
- Microsoft Certified Azure AI
- ECPC Finalist
- EOI Coaching
- AICE for AI
- DevOps Mastery Specialization